

Geometry-Aware Uncertainty Quantification for Frozen Vision–Language Models

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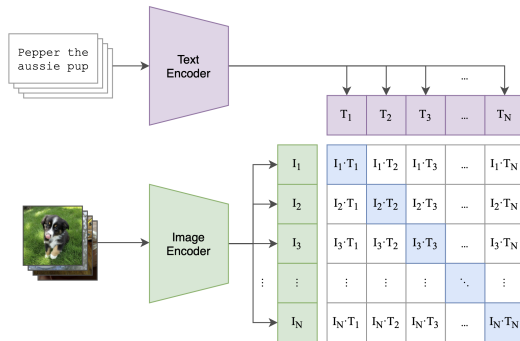
Dual-encoder vision–language models

Two encoders, one **shared representation space**¹ $\mathcal{Z}$: $f_I : \mathcal{I} \rightarrow \mathcal{Z}$, $f_T : \mathcal{T} \rightarrow \mathcal{Z}$.

Trained on image–caption pairs, against a similarity $s : \mathcal{Z} \times \mathcal{Z} \rightarrow \mathbb{R}$:

- matched pairs *pulled together*
- mismatched pairs *pushed apart*

Most models use cosine similarity as the metric $\Rightarrow$ Embeddings live on $\mathbb{S}^{d-1}$.



¹Radford, Kim, et al., ICML 2021; Zhai, Mustafa, et al., ICCV 2023.

How are they used

Deployed **frozen** and encoding once.

1. The embeddings are **semantic**

⇒ General-purpose *conditioning*:

- captioning
- visual question answering
- text conditioning for generative models

2. The two encoders are **aligned**

⇒ *Similarity lookup* across modalities:

- zero-shot classification
- cross-modal retrieval, at index scale
- filtering and curation of web-scale data

A **deterministic point** in, an embedding out: **No signal** of how much to trust it.

Uncertainty quantification

The Bayesian view splits predictive uncertainty into two additive terms²:

$$\underbrace{\text{Var}[y \mid x, \mathcal{D}]}_{\text{total}} = \underbrace{\mathbb{E}_{p(\theta|\mathcal{D})} [\text{Var}[y \mid x, \theta]]}_{\text{aleatoric}} + \underbrace{\text{Var}_{p(\theta|\mathcal{D})} [\mathbb{E}[y \mid x, \theta]]}_{\text{epistemic}}$$

Aleatoric: ambiguity in the data

What stays uncertain even once θ is known.
Irreducible — more data never removes it.

Epistemic: ignorance of the model

How much the parameters disagree on the output (conditioned on the data).
Reducible — better-targeted data does.

²Hüllermeier and Waegeman, Machine Learning 2021.

Where our works sit

How do we quantify uncertainty for a *single* pretrained model, **without posterior** over its parameters?

Our answer: drop the posterior, and build **distributions in embedding space** instead.

- aleatoric $\approx$ **spread of the match**: Does the query single out one counterpart, or many equally good ones?
- epistemic $\approx$ **embedding typicality**: Has the model seen embeddings like this before?

	Aleatoric	Epistemic	Both
Geometry-agnostic	GroVE, BayesVLM	MC Dropout	ProbVLM
Geometry-aware	AsymVLM	REPVLM	GeoFlowVLM

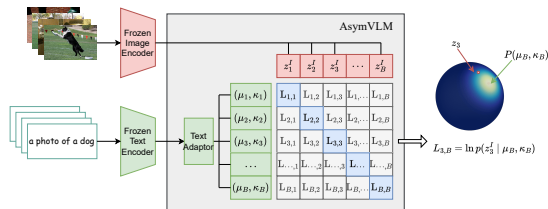
AsymVLM: quantifying aleatoric uncertainty³

Hypothesis. Retrieval is one-to-many not because images are ambiguous, but because **texts are**: “a dog” fits unboundedly many images.

Equip the **text** embedding with a **directional** distribution:

$$\mathbf{z}^T \sim P(\mu(t), \kappa(t))$$

which can be von Mises–Fisher or power spherical on $\mathbb{S}^{d-1}$.



Asymmetric by design: **text carries the variance**, image stays deterministic.

³ Ju, Andersson, et al., NeurIPS 2025.

Extending, not replacing, the CLIP objective

Let $\delta(r, s)$ denote the similarity between a text r and an image s . We have

$$\text{CLIP: } \delta_{\text{CLIP}} = \text{CosSim}(r, s)$$

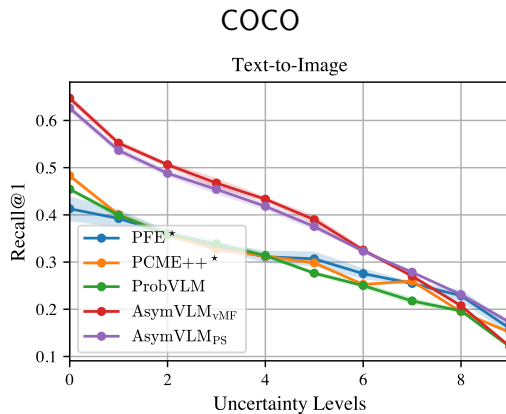
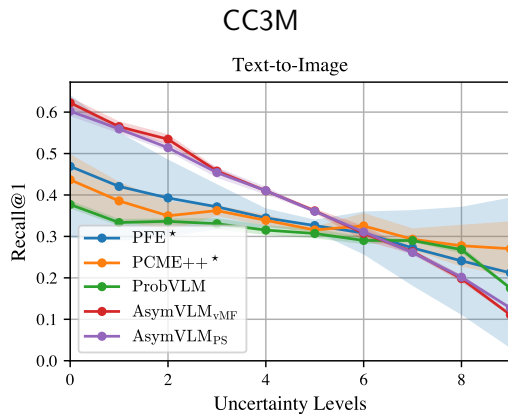
$$\text{AsymVLM}_{\text{vMF}} : \delta_{\text{vMF}} = \kappa(t_r) \cdot \text{CosSim}(r, s) + F_d(\kappa(t_r))$$

$$\text{AsymVLM}_{\text{PS}} : \delta_{\text{PS}} = \kappa(t_r) \ln(1 + \text{CosSim}(r, s)) + \ln C_d(\kappa(t_r))$$

Since $\kappa(t_r) \geq 0$, both are **monotonically increasing** w.r.t. δ_{CLIP} , with κ as an addition.

- An **extension of the CLIP objective**: μ models the semantic direction, while κ models the ambiguity of the text embedding.
- Inference extends beyond computing a nearest (geodesic) neighbor, to the **maximum-likelihood principle**.

Main results: uncertainty that tracks retrieval



Recall@1 against uncertainty level — ideally **monotonically decreasing**.

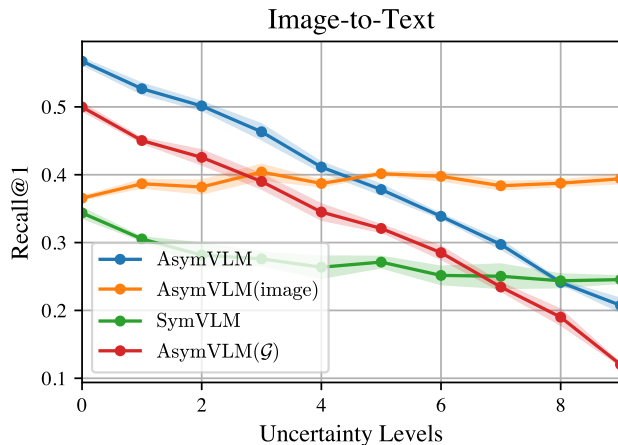
Ablation

AsymVLM($\mathcal{G}$)

Text is mapped to a Gaussian in $\mathbb{R}^d$ and then normalised, with a Gaussian kernel in the loss.

- Uncertainty survives.
- Retrieval does not.

The asymmetry buys calibration, while geometry buys performance.



REPVLM: quantifying epistemic uncertainty⁴

The Bayesian definition needs a posterior

$$U_{\text{ep}}(x) = \text{Tr} \left(\text{Cov}_{p(\theta|D)}(f(x; \theta)) \right)$$

No posterior access to frozen, billion-parameter encoders.

Data typicality: replace it with the input density $-\log p(x)$

Density is dominated by low-level statistics, not semantics.

Representation typicality: measure it after the encoder

$$U_{\text{ep}}(x) = -\log p(f_{\theta}(x))$$

Measure typicality where the semantics live.

⁴ Ju, Nautiyal, et al., ICML 2026.

Representation Density estimation

Goal. Estimate $p(z)$ for $z \in \mathbb{S}^{d-1}$ that is

- **high-dimensional** (hundreds of dimensions),
- **multi-modal** (concepts cluster across the sphere),
- **respects the spherical geometry**.

Options

- Parametric directional distributions – too simple, can't capture multimodality.
- KDE / GMM – collapse in high dimensions, no geometry.
- **Riemannian flow matching (RFM)** – scales to high dim, captures multimodality, and operates **natively on the sphere**.

Building RepVLM

Training – learn the vector field.

- 1 Design a probability path p_t from $p_0 = \text{Uniform}(\mathbb{S}^{d-1})$ to $p_1 = p(z \mid c)$, generated by a velocity field $u_t(z, c)$ via the ODE

$$\dot{z}_t = u_t(z_t, c), \quad z_t \in \mathbb{S}^{d-1}, \quad u_t(z_t, c) \in T_{z_t} \mathbb{S}^{d-1}, \quad c \in \{\text{text, image}\}$$

whose trajectories follow **geodesics** of the sphere.

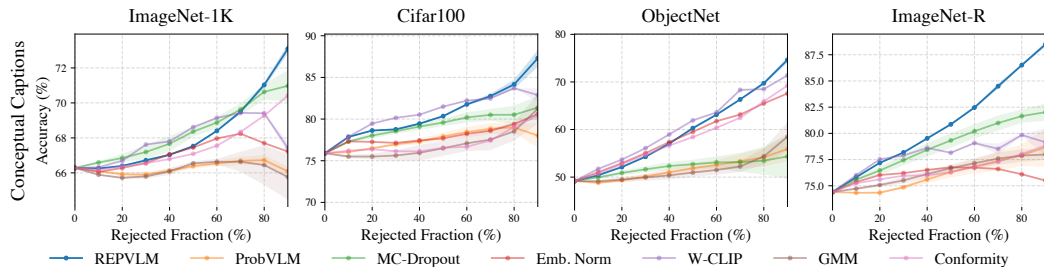
- 2 Regress u_t with a neural network $v_t(z_t, c; \phi)$:

$$\min_{\phi} \mathbb{E}_{t, z_t, z_1, c} \|v_t(z_t, c; \phi) - u_t(z_t, c \mid z_1)\|^2.$$

Inference – evaluate density.

Integrate $\nabla \cdot v_t(\cdot; \phi)$ along the geodesic flow from $t = 0$ to $t = 1$ to recover $\log p_1(z \mid c)$ **exactly**.

Main results

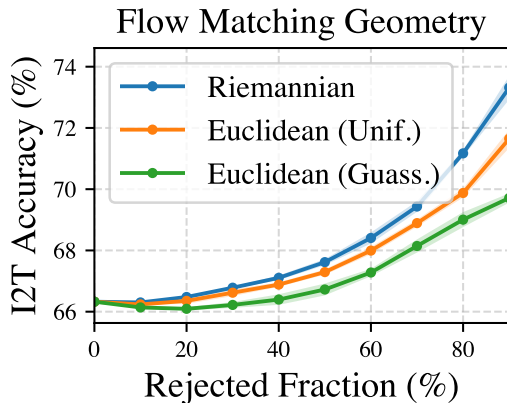


- **Monotonicity** between rejection rate and prediction error.
- Effective for **OOD detection** and **automated data curation**.
- Only $\sim 10\%$ the compute of MC-Dropout at inference.

Geometry ablation

- 1 **Riemannian**: $P_0 = \text{Uniform}(\mathbb{S}^{d-1})$, transported along **geodesics**.
- 2 **Euclidean (Unif.)**: $P_0 = \text{Uniform}(\mathbb{S}^{d-1})$, transported along **straight lines** through $\mathbb{R}^d$.
- 3 **Euclidean (Gauss.)**: $P_0 = \mathcal{N}(0, I)$, along straight lines.

One ingredient at a time: $1 \rightarrow 2$ swaps the **paths**, $2 \rightarrow 3$ swaps the **base**.

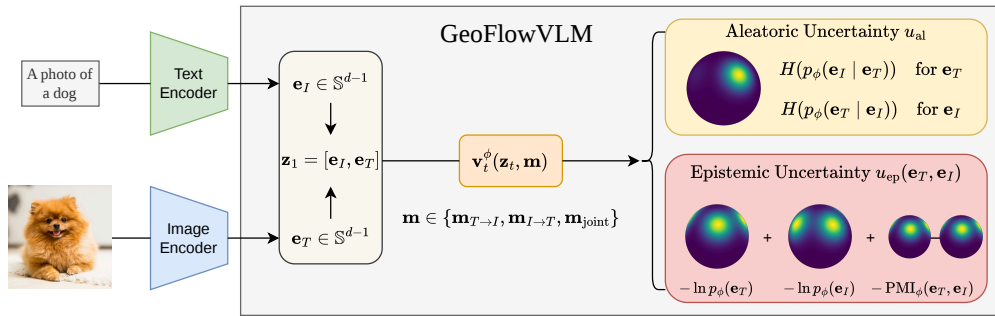


GeoFlowVLM: learn the whole distribution⁵

AsymVLM learns **conditionals** and REPVLM learns **marginals**.

Model the **whole** distribution of both text and image embeddings.

⇒ Both can be read out from one model.



⁵Nautiyal, Ju, et al., arXiv 2026.

Training: cross-modal inpainting

The idea. Treat the pair $\mathbf{z} = [\mathbf{e}_I; \mathbf{e}_T] \in \mathcal{M}$ as *one* object, and train as in **inpainting**: hold an arbitrary part of $\mathbf{z}$ fixed, and flow the rest.

A **single** velocity field $\mathbf{v}_t^\phi(\mathbf{z}_t, \mathbf{m})$, conditioned on a block mask $\mathbf{m} \in \{0, 1\}^{2d}$:

- $m_k = 1$: dimension is *transported* by the flow
- $m_k = 0$: dimension is *held* at its observed value, its stream's time pinned to 0

$$\mathbf{m}_{\text{joint}} = [\mathbf{1}; \mathbf{1}] \quad \mathbf{m}_{I \rightarrow T} = [\mathbf{0}; \mathbf{1}] \quad \mathbf{m}_{T \rightarrow I} = [\mathbf{1}; \mathbf{0}]$$

Supervision is masked to the generated dimensions, along *geodesics* on each factor:

$$\mathcal{L}(\phi) = \mathbb{E}_{t, \mathbf{z}_0, \mathbf{z}_1, \mathbf{m}} \left\| \mathbf{m} \odot (\mathbf{v}_t^\phi(\mathbf{z}_t, \mathbf{m}) - \mathbf{v}_t(\mathbf{z}_t \mid \mathbf{z}_1)) \right\|_{g_{\mathbf{z}_t}}^2$$

Inference: reading the uncertainties out

All three log-densities come from the *same* model, by reverse-time integration.

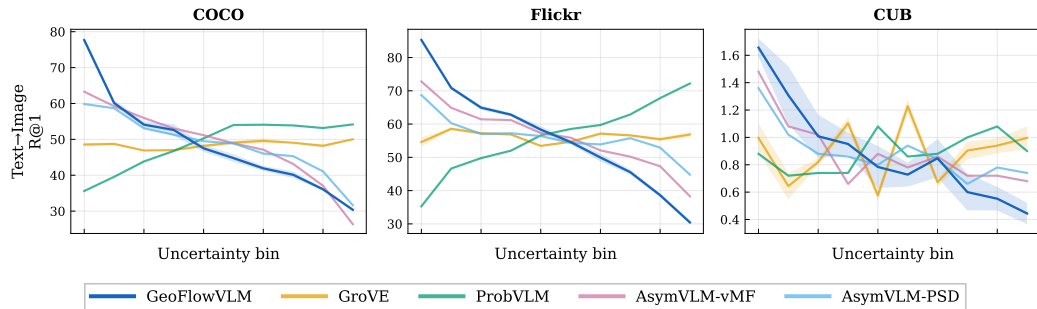
Aleatoric — entropy of the conditional

Sample from $p_\phi(\mathbf{e}_T \mid \mathbf{e}_I)$ and report the entropy estimate.

Epistemic — the chain rule splits two different signals

$$-\log p_\phi(\mathbf{e}_I, \mathbf{e}_T) = \underbrace{\left[-\log p_\phi(\mathbf{e}_I) - \log p_\phi(\mathbf{e}_T) \right]}_{\text{typicality: coverage of the region}} \underbrace{- \text{PMI}_\phi(\mathbf{e}_I, \mathbf{e}_T)}_{\text{matching: confidence-like}}$$

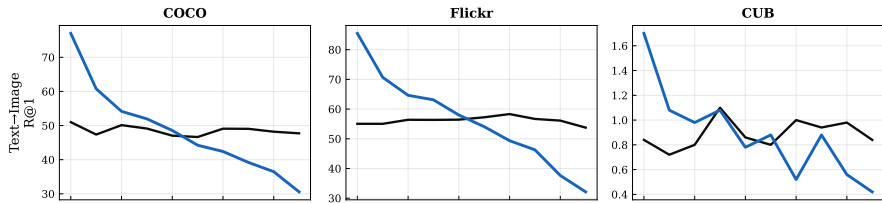
Main results



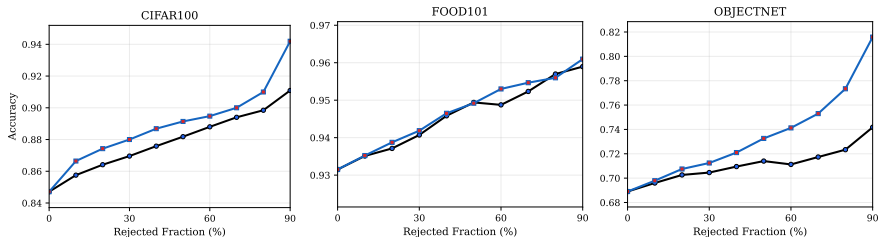
Disclaimer: We do **not** claim that GeoFlowVLM out performs AsymVLM or REPVLM theoretically. In practice, GeoFlowVLM outperforms mainly due to **better estimation**.

Geometry ablation

Aleatoric



Epistemic



—●— Euclidean —■— Geometric

Takeaways

A frozen encoder turns uncertainty into density estimation

With no parameter posterior available, **representation density** is a power surrogate.

Geometry is a correctness condition

- Theoretically, geometry is a **hard constraint** induced by cosine similarity.
- Empirically, the Euclidean controls **lose** consistently.

Thank you for your attention!

Questions?

To know more

